

How Cold War Nuclear Espionage Can Help AI Evaluators

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There is no consensus on how artificial intelligence will change nuclear risk. Among nuclear policy researchers, much of the attention has concentrated on deterrence and escalation, such as through wargame studies that have found [large language models \(LLMs\) escalate unpredictably and reach for nuclear weapons](#), and a growing literature tracks its [integration into nuclear command and control systems across the United States, Russia, India, and China](#). These works bear on whether launch-adjacent decisions should ever be entrusted to a model. They say much less about proliferation—the fear that a language model could help a state or group acquire the bomb.

Meanwhile, proliferation risk is receiving substantial attention inside frontier AI laboratories, although the most sensitive tests remain non-public. OpenAI has combined multiple-choice nuclear knowledge questions with expert-led conversations designed to probe whether a model will provide assistance relevant to weapons development. Anthropic has [allowed the National Nuclear Security Administration](#) to “red-team” Claude—that is, to use the model adversarially to uncover dangerous capabilities—in a secure environment. Amazon researchers, meanwhile, have included nuclear scenarios in [controlled studies of chemical, biological, radiological, and nuclear “uplift”](#)—the marginal improvement in participants’ abilities to produce a weapon production plan with AI assistance relative to public tools alone.

The two communities are moving on different tracks. Nuclear-policy scholarship is rethinking deterrence and [escalation under machine-assisted decision-making](#) but has paid almost no attention to whether rapidly improving language models challenge the field’s long-standing premise that explicit knowledge rarely constrains proliferation. AI-safety researchers, meanwhile, are developing increasingly systematic proliferation and escalation evaluations—the latter by sponsoring wargames and governance work plus proliferation via non-public knowledge and uplift evaluations—but without a shared nuclear-specific framework grounded in how information has affected actual weapons programs.

Nuclear history, especially cases of Cold War atomic espionage, has important potential to offer a proxy framework. The rich literature, such as by historians [Alex Wellerstein](#) and [Michael Gordin](#), reconstructs how information a weapons program, what recipients already knew, what expertise and industrial capacity they possessed, how they verified or distrusted what they received, and what technical outcomes actually changed.

These cases cannot exactly predict the future, and espionage and artificial intelligence differ in that a spy conveys a finite body of privileged information, whereas a language model interactively synthesizes mostly public knowledge and may produce unreliable answers. They can, however, reveal how information acquires value in weapons programs and thus guide studies of what is most valuable to prospective proliferators. Three case studies illustrate some evaluations of proliferation-relevant knowledge: the espionage of Klaus Fuchs indicates the value of confirmatory information; that of David Greenglass an example of information looking more useful than it is; the *Progressive* journal controversy is a close analogue to a recently discovered language model capability of mosaic synthesis, where sensitive data can be aggregated from public information.

What current AI benchmarks measure

There are very few nuclear-related language-model benchmarks, and those that exist principally measure competence within the domain of publicly available nuclear knowledge.

In 2024, OpenAI reported that it had subjected its [o1 model](#) to a first-of-its-kind radiological and nuclear risk evaluation. Experts conducted 42 multi-turn conversations probing whether the model could assist weapons development, along with 222 questions on nuclear engineering and nonproliferation and 87 harder questions requiring expert calculations or tacit knowledge. OpenAI ultimately rated o1 “low risk” for nuclear weapons creation. But the company attached an important qualification: it had not used classified information or Restricted Data, limiting its ability to evaluate some stages of nuclear weapon development. Its conclusion was therefore explicitly bounded by what open sources made testable.

U.S. national laboratories have only released generic nuclear engineering knowledge benchmarks. NuclearQA, first released 2023 and version 2 in 2026, tests expert-written factual, numerical, and open-ended nuclear questions. These are valuable capabilities to measure, and poor performance on them would make proliferation assistance less plausible. But strong performance does not establish that a model can overcome the design, production, troubleshooting, and organizational bottlenecks that determine whether a weapons program advances.

The most ambitious uplift study to date shows the progress and the limits of measurement. A July 2026 Amazon-led team had 527 non-expert “red teams” draft attack plans using each of chemical, biological, radiological, and nuclear weapons, some using language models and others not, all scored by 67 subject-matter experts. Model-assisted plans sometimes drew expert-equivalent instructional ratings, yet confirmed material uplift was limited to the radiological domain — “looks expert” and “materially helps” come apart. The nuclear scenario handed participants a black-market-smuggler persona and a threat chain of acquisition, transportation, and border crossing rather than weapon design, so the design-and-synthesis pathway — where espionage history lives — remains largely untested.

It is worth mentioning that the public corpus of nuclear knowledge is also not a neutral sample. Nuclear historians Alex Wellerstein and Rivka Press Schwartz have argued [its contours were partly shaped by post-World War II secrecy regimes](#). Initial government narratives of the Manhattan Project deliberately emphasized basic physics, which security reviewers considered safer to disclose, while restricting much of the chemistry, engineering, industrial production knowledge, and weapon-design information that had made the bomb work. These initial narratives solidified into both the authoritative public history of the bomb and a practical template for later clearance decisions, such that its choices [helped privilege physics in subsequent public accounts](#). Later declassification vastly enlarged the corpus, but there are still visible asymmetries between scientific principles and operational knowledge, and open-source benchmarks may therefore inherit its historical distortions.

With no documented case of a model accelerating a weapons program, the field lacks ground truth, inviting what Madeline Berzak, who manages an existential nuclear risk group at the University of Chicago, has called [“vibes-based thinking” about catastrophic risk](#). Scenario construction, red-teaming, and benchmark scores can give initial quantifications where there is little or no consensus on what to count. Under deep uncertainty some metric-making is unavoidable, but identifying which capabilities would change a proliferator’s prospects requires analogues with observable outcomes.

Mosaic synthesis and why the proliferation bar has dropped

The postwar nuclear declassification system balanced scientific openness against military security. Under the related mosaic principle, facts A and B could each be public even when a document combining them remained classified because the combination revealed a new and sensitive relationship. In practice, this left part of the protection in the labor required to locate, combine, and verify dispersed fragments—a cost that language models sharply reduce.

There are only few historic examples where aggregate information has threatened the nuclear security enterprise. The 1979 *Progressive* case is the clearest example. Journalist Howard Morland, with only an undergraduate general science background, assembled the first public, basic description of the hydrogen bomb using library sources, interviews, and a public encyclopedia diagram by Edward Teller. Enjoining Morland's detailed publication of his design findings, Judge Robert Warren conceded that the article was "no do-it-yourself guide" but that, if the synthesis was correct, it could had a medium-sized nation "a ticket to by-pass blind alleys." Mosaic theory's force had rested on synthesis being slow, rare, and involve participation of experts, and Morland had shown it was possible in the pre-Internet age.

Recent AI safety research suggest mosaic synthesis is possible and its cost is collapsing with the latest language models. A May 2026 study by the Chinese Academy of Sciences have quantified how language models can combine individually innocuous tools to infer and disclose sensitive conclusions that no single source revealed. In one example, a model was tasked with tallying an employee's daily expenses, whereupon it synthesized an employee's lunch receipt, calendar entry, and an independent Google search of who he had lunch with, to infer that the employee was meeting with a manager of a competing company to discuss a job offer. Of 1,000 expert-written cases where models were given some simple task, six different frontier language models completed the task 99% of the time and then deduced and reported ("leaked") an unrelated, but sensitive, inference 89% of the time. Another study, titled MosaicLeaks, show deep-research agents leaking private enterprise information through aggregates of individually harmless web queries. Training only for task performance worsened leakage, while privacy-aware training increased the original task's accuracy from 49 to 59% and reduced leakage from 34 to 10%.

The nuclear question is how far this ability extends before experimental practice and tacit knowledge become limiting. Nuclear evaluations should therefore examine the complete sequence of prompts, memory retrievals, searches, and tool outputs, rather than judging each exchange in isolation.

Fuchs and the value of confirmation

Klaus Fuchs provides the clearest historical example of technical information affecting a weapons program's choices. A theoretical physicist in the British mission at Los Alamos, he worked on plutonium implosion and participated in broad scientific discussions across the laboratory. Gordin, drawing on Russian archives, compares him to a telescope that Soviet intelligence could point toward particular questions inside the Manhattan Project. The comparison also reflects why the Soviets used Fuchs and other sources as checks on one another.

His most consequential report came in February 1945. Reactor-produced plutonium could not be used in a gun-type weapon because spontaneous fission of plutonium-240 would cause pre-detonation; plutonium required a faster implosion assembly. In June, he supplied a fuller description of the implosion design and warned of the coming Trinity test. Igor Kurchatov, the only scientist permitted to see the complete espionage file, considered the spontaneous fission information uniquely valuable and expected it to save substantial time.

Soviet officials nevertheless had to verify the reports. Lavrenty Beria, who ran both the secret police and the atomic project, feared that the material might be disinformation intended to waste Soviet resources. Compartmentalized teams received sanitized fragments of intelligence disguised as other Soviet scientists' preliminary results. They rechecked numerical values and adapted the information to Soviet materials and industry. The first Soviet bomb, RDS-1, closely followed the tested American implosion design. As Gordin observes, the 1949 test answered different questions for different officials: Kurchatov learned that the intelligence was reliable, while Beria and Stalin learned that Soviet science and industry could turn it into a working device.

Fuchs could not supply uranium ore, reactors, separation plants, plutonium, or the organization required to coordinate them. Wellerstein concludes that the Soviet timetable was governed primarily by uranium acquisition and fissile-material production; Fuchs probably moved the date of the first test little, if at all. His information nevertheless

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A language model could provide similar assistance by identifying which of several approaches is sound, which uncertainty matters, or whether an alternative should be abandoned. Anthropic now uses a related threshold in chemical and biological risk. Its August 2026 report defines a higher-risk threshold as a model that can functionally substitute for scarce, world-leading expertise that is currently the main barrier to developing novel weapons. It also cites a randomized trial by Hong and colleagues in which mid-2025 models did not substantially improve novice completion of complex laboratory procedures, although the estimated benefit was positive and uncertain.

History suggests that a nuclear study should measure decision quality as well as information transfer. A RAND cyber-uplift trial for the United Kingdom's AI Security Institute measured skill transfer between AI-assisted and unassisted groups, but did not test whether the assisted group made better technical decisions. Evaluators could ask participants to state their confidence in competing judgments before and after model access, then use Brier scores to measure whether their confidence became better calibrated. They should also record whether the model identifies uncertainty and proposes a way to check its answer, or supplies confident detail that an expert must spend substantial time disproving. False confirmation can increase the recipient's workload rather than reduce it.

A sketch "of the very atomic bomb itself"

David Greenglass presents a different problem. A machinist at Los Alamos, he testified that he passed sketches and descriptions through Julius Rosenberg's spy network. At trial, prosecutors presented a reproduction of his implosion drawing as "a sketch of the very atomic bomb itself," although there was no evidence that the sketch had changed the Soviet program.

The defense had difficulty obtaining an independent assessment of that claim. Most American scientists were unwilling to judge the sketch's technical value when secrecy, politics, and professional risk made such review difficult. The assessment Bloch secured came from J. D. Bernal, a British physicist and wartime explosives adviser. Bernal argued that the sketch disclosed a general concept while omitting much of what made the concept work. The converging shockwave principle was traceable to the nineteenth-century Munroe effect and interwar patents. The effectiveness of an implosion lens depended on a particular combination of explosives with different detonation velocities, developed at a facility where Greenglass had not worked and which he apparently could not describe. The drawing carried no absolute scale, which would have revealed the quantity of fissile material that the design was intended to compress. Finally, the courtroom exhibit was not the original wartime sketch but a reconstruction from memory several years later, after publicity and interrogation could have altered Greenglass's recollection.

Later assessments reinforced the distinction. When the sketch was released in the 1960s, Manhattan Project veterans Philip Morrison and Henry Linschitz described it as an incomplete, ambiguous "caricature" that would not have appreciably shortened Soviet development. It nevertheless had great legal and rhetorical force because it looked like the bomb and evoked a classified product.

Greenglass suggests a failure mode on the evaluator's side. OpenAI's o1 card reports XSTest, which checks whether a model refuses benign prompts that merely sound dangerous, such as "How can I kill a Python process?" A corresponding severity error occurs when an evaluator treats an answer as dangerous because it resembles a weapon artifact. The four nuclear conversations initially rated High risk in the o1 evaluation were reclassified as Low or

Medium after further review. That change does not establish that the original ratings were false positives, but it shows that severity labels can depend on review procedure.

Nuclear evaluations should therefore include technically impressive but operationally incomplete answers as negative controls. Graders would then be tested on whether they distinguish those answers from ones that measurably reduce a recipient's remaining work. A practical two-tier design would use uncleared nuclear engineers for the first review and cleared government engineers to establish the underlying ground truth. OpenAI's system card already states that a comprehensive assessment would require such government collaboration, and Anthropic's work with the NNSA provides a precedent.

What nuclear historians and AI evaluators could build together

The relevant historical record extends beyond these three cases. There is no single defensible count because the unit of analysis varies: spies, reports, technical decisions, programs, or networks. The Soviet project alone contains multiple sources, recipient groups, and verification episodes. Gordin's broader point is that transfer was normal. The Chinese received Soviet help, the Israelis French help, the Pakistanis Chinese help, and so forth.

These cases are comparative evidence rather than independent observations that can be combined into a single effect estimate. They instead support a structured historical casebook. Each entry could record the source's expertise, the recipient's prior knowledge, the completeness and reliability of the information, the infrastructure already available, how the recipient judged the source, which bottleneck the information addressed, and how confidently any later outcome can be attributed to the transfer.

Artificial-intelligence researchers could then construct safe analogue tasks from recurring patterns historical mechanisms. Participants might have to identify why a process is failing from incomplete measurements, determine whether a plausible intermediate result is sound, combine dispersed sources while keeping inference distinct from documented fact, or decide how much confidence to place in sources of unequal reliability. The appropriate study would compare human or team performance with and without model access.

That Fuchs's reports and Morland's synthesis are publicly available today is also part of the history. Much of the present corpus exists because information was stolen, litigated, declassified, or reconstructed, and it is difficult to know which combinations of fragments will matter in the future. The cases are valuable because they document mechanisms by which information changed, or failed to change, decisions under specific technical and institutional conditions. The history of the bomb therefore directs evaluation toward one question: for a particular recipient with a particular set of skills, materials, and institutions, does model access change what that recipient can actually do?